LEXTRIA RESEARCH

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12-10-2026

DUE DILIGENCE REPORT: AR1115

Dear Inventor,

We have completed our due diligence under the Patents Act, 1970 and the Computer Related Inventions Guidelines issued by the Hon. Indian Patent Office, as instructed by Hon. IPTTO MAHE. We set out our observations in respect of the invention titled Deterministic Multimodal Fusion Architecture for Zero-Day Phishing Detection, Computational Optimization and Cryptographic Feature Provenance, being a computer-implemented system that evaluates the entropy of a uniform resource locator, selectively bypasses HTML parsing and page rendering, and records a provenance trail for each feature used in the decision. We refer to the 31-page Technical Specification as the disclosure and to the supplied source code as the code. We would be grateful for your help on the points below.

1. SECTION 3(k) OBSERVATIONS: ENTROPY GATE, STRUCTURAL METRIC, FUSION AND PROVENANCE

We would like to kindly share a few observations regarding Section 3(k) of the Patents Act, 1970, read with the Computer Related Inventions Guidelines 2025, module by module.

- Lexical entropy gate (disclosure, page 5, FIG. 2; page 10, item 5.1; page 12, FIG. 9): The comparison of H(U) with the threshold of 4.50 bits per character (page 10, item 5.6) presently reads as a mathematical evaluation. It would be helpful if the measured reduction in parsing and rendering work, under stated conditions, could kindly be shared, since the count of bypass events (page 16, Case E) is not, as the disclosure kindly notes, a measure of processor cycles saved.

- Structural density ratio (page 6, FIG. 3; page 10, item 5.2): The ratio of node count to maximum tree depth is a mathematical expression, and we would be grateful if the

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technical consequence of the 50% deviation threshold and of the harvester signature at page 10, item 5.6, could kindly be described.

- Fusion policies (page 7, FIG. 4; pages 11 to 14, items 6.1 to 6.6): The five strategies are presently stated as formulae and a lookup, and it would be helpful to have the technical consequence of the reference weights of 0.40, 0.40 and 0.20.

- Provenance and verification (page 7, FIG. 5; page 14, item 7.1): This is the module closest to a technical process, and it would be helpful if the canonical encoding, the fields included within the hashed bytes and the verification step could kindly be described.

We have identified three areas where your kind clarification would help

- Mathematical Method (para 4.5.1): It would be helpful to describe a technical consequence independent of the computation, such as measured processing time, memory and latency per input with and without the entropy gate.

- Algorithm (para 4.5.3): It would be helpful to have the named components, interfaces and data structures for the gate, the structural extraction and the fusion step, including the rule by which a bypassed modality is recorded.

- Computer Programme Per Se (para 4.5.4): It would be helpful to have measured results for the effect of the gate on detection quality, so that the saving in work can be shown alongside an unchanged or acceptable detection outcome.

Assistance needed from inventor: We would be grateful if you could kindly help us with the following: (i) for the entropy threshold of 4.50, the deviation threshold of 50%, the fusion weights and the outcome risk values of 0.99, 0.95, 0.70 and 0.05 (pages 10 and 11, item 5.6), the reason for each value and the consequence of departing from it; (ii) the late-fusion decision threshold and the early-fusion projection weights (page 11, item 5.6); (iii) a comparison of the pipeline with and without the entropy gate on the same labelled set of URLs and pages, reporting processing time, memory, latency and detection accuracy; and (iv) the Guidelines paragraph relied on, if a different characterisation is preferred.

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2. SECTION 3(m) OBSERVATIONS: RULE-BASED DECISION LOGIC

We would like to kindly share a few observations regarding Section 3(m) of the Patents Act, 1970, which excludes a mere scheme or rule or method of performing a mental act.

- Gated decision rules (page 11, Table 6-1; page 12, FIG. 9): The sequence of comparing entropy with a threshold, then a structural deviation with a further threshold, and then the TLS state, could presently be read as a set of rules that a person might apply manually. It would be helpful to know how these rules control the operation of the parser and renderer within the system.

- Consensus voting and Cartesian risk matrix (page 13, items 6.4 and 6.5): The two-of-three vote and the eight-cell lookup are presently described as a scheme, and we would be grateful for the manner of their execution on the stored feature records and the decision for each cell of Table 6-2.

Assistance needed from inventor: We would be grateful if you could kindly describe how each decision rule acts on the system itself, for example by withholding the start of HTML parsing or page rendering, and the resulting change in processing load.

3. SECTION 3(n) OBSERVATIONS: EXPLANATION AND REPORT OUTPUT

We would like to kindly share a few observations regarding Section 3(n) of the Patents Act, 1970, which excludes the presentation of information.

- Explanation templates (page 8, FIG. 6; page 15, item 7.3): The human-readable explanation string presently reads as a manner of presenting the decision. It would be helpful to know how the explanation is bound to the recorded branch and provenance digests, so that it serves a verification function.

- Evidence report (page 15, item 7.4): The JSON report presently describes the content and layout of an output, and we would be grateful for the manner in which the report is used to verify or replay a decision.

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Assistance needed from inventor: We would be grateful for a description, with an example, of how the explanation and the report are verified against the stored feature records, and the result when a recorded value is altered.

4. SECTION 10(4) SUFFICIENCY AND ENABLEMENT: CLARIFICATIONS

We would be grateful for your kind clarification on the following points, so that the description of the invention remains complete and consistent.

- Module implementation: The disclosure, page 17, item 9.1, refers to test_prior_art_inventions.py and examples.py, which were not part of the supplied archive. We would be grateful for these, together with the working versions of the structural density ratio computation, the SHA-256 provenance record, the orchestrator, the lineage tracker and auditor (page 15, item 7.2), the explanation module and the TLS check.
- Gated policy and missing evidence: Kindly confirm whether the password input and the maximum depth of four or less (page 10, item 5.6) are applied before the PHISHING_CLONE or SUSPICIOUS outcome (page 12, FIG. 9), and describe the rule applied when HTML, TLS or any other modality is unavailable or bypassed, including consensus voting with fewer than three modalities (page 13, items 6.2 to 6.4).
- Input scope and parsing profile: Kindly confirm whether entropy is computed over the entire URL, the hostname or the registrable domain, with the normalisation applied (page 11, item 5.6), and which node types are counted, together with the source of the brand baseline (page 10, item 5.2).
- Test results: Kindly share the logs behind the results at page 18, Table 9-2, and the sample behind the figure of 3.80 bits per character for legitimate domains (page 10, item 5.6), so that they may be placed on record.

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5. CONSOLIDATED SUPPORT WE WOULD BE GRATEFUL TO RECEIVE

Comparative performance data (supporting Parts 1 and 2)

Processing time, memory, latency, precision, recall and false-positive rate for the pipeline with and without the entropy gate, on the same labelled URLs and pages, would be greatly appreciated.

Parameters and their basis (supporting Parts 1 and 2)

We would be grateful for the reason behind each reference value at pages 10 and 11, item 5.6, the effect of a higher or lower value, the late-fusion decision threshold, the early-fusion projection weights and the decisions for the Cartesian risk matrix.

Clarifications (supporting Parts 1 to 4)

Your kind replies to Parts 1 to 4, including the modules and logs at Part 4 and the example at Part 3, would be most appreciated.

Should the technical details requested above be unavailable at this stage, the software code and the associated documentation may be protected as original works under the Copyright Act, 1957.

Until the above clarifications are received and the additional information is placed on record, our observations under Sections 3(k), 3(m) and 3(n) of the Patents Act, 1970 will remain open. Given the absence of the supporting comparative data identified above, we recommend proceeding by way of a Provisional Specification at this stage to secure a priority date, with the Complete Specification to follow once the above material is placed on record. We will finalise our recommendation on the final claim scope, and proceed with drafting the Complete Specification, upon receipt of the above.

Regards,

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